

House Price Prediction - Project Summary

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Problem Statement

Both real estate sellers and buyers usually estimate the value of their property with the help of intuition and comparison method, not with the help of data-based approach. In this project, the objective is to create a regression model for predicting the prices of houses based on their features (like area, number of rooms, location facilities, and furnishing) and determine which of those features have the strongest impact on the price.

Dataset & Approach

Housing dataset with 545 observations of 13 attributes including area, bedrooms, bathrooms, stories, parking, and several binary features (availability of a main road, guest room, basement, hot water heating, air conditioning, and preferred locality) as well as furnishing attribute has been used in this analysis. There was no missing or duplicated data in the dataset. Categorical yes/no and furnishing variables were transformed into numeric format with one-hot encoding before modelling. The data was split into 80% for training and 20% for testing datasets, and two regression models were built.

Key Findings

Area, number of bathrooms, and presence of air conditioning were found to be among the most highly correlated features with the price, with area being the dominant one by itself. Stories, parking space, and preferred location were positively correlated features with price.

Model Performance

The R^2 score for the Linear Regression model was found to be approximately 0.65, which means that it explains about 65% of variance of house prices with an average MAE of about ₹9.7 lakh. Random Forest did not do as well as Linear Regression ($R^2 \approx 0.61$) and that may come as a surprise due to the limited amount of data available (545 entries).

Conclusion & Recommendation

From the model results, it is evident that size and important amenities play a much more critical role in determining price than other factors like number of bedrooms, for example. This means that a real estate business must consider focusing more on such amenities in their pricing strategy and advertisement.